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(1) $G: 10.5 \text{ kV}$

$T1: 10.5 \text{ kV} / 121 \text{ kV}$

$T2: 110 \text{ kV} / 38.5 \text{ kV} / 11 \text{ kV}$

$T3: 35 \text{ kV} / 6.6 \text{ kV}$

$T4: 10 \text{ kV} / 0.4 \text{ kV}$

$D: 6.3 \text{ kV}$

$L: 380 \text{ V}$

(2) $T1: 10.5 \text{ kV} / 124.025 \text{ kV}$

$T2: 110 \text{ kV} / 38.5 \text{ kV} / 11 \text{ kV}$

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- 1) 根据 $S_{12} = \frac{S_2(Z_{23}^* + Z_{31}^*) + S_3 Z_{31}^*}{Z_{12}^* + Z_{23}^* + Z_{31}^*}$ 算出 S_{12}
- 2) 根据基尔霍夫电流定律分别求出其他支路功率初步分布,找到功率分点
- 3) 在无功分点处将网络解开成辐射网
- 4) 设全网为额定电压,从末端负荷节点向1号节点推算各支路功率
- 5) 从1节点用给定的电压和已算出的功率向末端推算各节点的电压
- 6) 再用给定功率和算得的电压从末——首推算功率和电压。直到满足首端电压和末端功率为止

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$$\begin{cases} \dot{U}_1 - I_1 Z_T = k U_2 \\ \dot{I}_2 = -k \dot{I}_1 \end{cases}$$

$$\Rightarrow \begin{cases} y_{12} = \frac{Z_T}{k} \\ y_{10} = \frac{Z_T}{1-k} \\ y_{20} = \frac{Z_T}{k(k-1)} \end{cases}$$

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$$(1) \quad k_A = k_{GA} + k_{LA} = 850 \text{ MW/Hz}$$

$$k_B = k_{GB} + k_{LB} = 1100 \text{ MW/Hz}$$

$$\begin{cases} 100 + \Delta P_{AB} = -850 \Delta f \\ 50 - \Delta P_{AB} = -1100 \Delta f \end{cases}$$

$$\text{解得} \begin{cases} \Delta P_{AB} = -34.62 \text{ MW} \\ \Delta f = -0.077 \text{ Hz} \end{cases}$$

$$(2) \quad k_A = k_{GA} + k_{LA} = 850 \text{ MW/Hz}$$

$$k_B = k_{LB} = 400 \text{ MW/Hz}$$

$$\begin{cases} 100 + \Delta P_{AB} = -850 \Delta f \\ 50 - \Delta P_{AB} = -400 \Delta f \end{cases}$$

$$\text{解得} \begin{cases} \Delta P_{AB} = 2 \text{ MW} \\ \Delta f = -0.12 \text{ Hz} \end{cases}$$

A,B两系统由联络线相联,两系统发电机组的单位调节功率为: $K_{GA}=800\text{MW/Hz}$, $K_{GB}=700\text{MW/Hz}$;负荷调节效应 $K_{LA}=50\text{MW/Hz}$, $K_{LB}=400\text{MW/Hz}$;两系统的负荷增量为。求在下列情况下系统频率变化量和联络线功率的变化。(10分)(1)当A、B均参加一次调频;(2)当A参加一次调频,而B不参加一次调频。

B I U   手机传图 Σ 代码语言 

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2.

$$\Delta U_{Tmax} = \frac{28 \times 2.1 + 21 \times 38.5}{112.09} = 7.74 \text{ kV}$$

$$\Delta U_{Tmin} = \frac{15 \times 2.1 + 10 \times 38.5}{115.45} = 3.61 \text{ kV}$$

采用逆调压: $U_{2max} = 10.5 \text{ kV}$, $U_{2min} = 10 \text{ kV}$

$$U_{1tmax} = (112.09 + 7.74) \times \frac{10.5}{10.5} = 119.83 \text{ kV}$$

$$U_{1tmin} = (115.45 + 3.61) \times \frac{10.5}{10} = 125.01 \text{ kV}$$

$$U_{1tav} = \frac{1}{2} (U_{1tmax} + U_{1tmin}) = 122.42 \text{ kV}$$

故选择 主 分接头, 即 $U_{1t} = 121 \text{ kV}$

校验低压侧母线电压:

$$U_{2max} = (112.09 + 7.74) \times \frac{10.5}{121} = 10.40 \text{ kV} < 10.5 \text{ kV}$$

$$U_{2min} = (115.45 + 3.61) \times \frac{10.5}{121} = 10.33 \text{ kV} > 10 \text{ kV}$$

满足要求

3.

$$\tilde{S}_2 = P_2 + jQ_2 = 20 + j15 \text{ MVA}$$

$$\begin{aligned}\Delta \tilde{S}_{Y2} &= -jBU_2^2 = -j2 \times 10^{-5} \times 110^2 \\ &= -j0.242 \text{ MVar}\end{aligned}$$

$$\tilde{S}'_2 = \tilde{S}_2 + \Delta \tilde{S}_{Y2} = 20 + j14.758 \text{ MVA}$$

$$\begin{aligned}\Delta \tilde{S}_L &= \frac{P_2'^2 + Q_2'^2}{U_2^2} (R + jX) \\ &= \frac{20^2 + 14.758^2}{110^2} \times (3 + j50) \\ &= 0.153 + j2.553 \text{ MVA}\end{aligned}$$

$$\tilde{S}'_1 = \tilde{S}'_2 + \Delta \tilde{S}_L = 20.153 + j17.311 \text{ MVA}$$

$$\Delta U_2 = \frac{P_2' R_L + Q_2' X_L}{U_2} = \frac{20 \times 3 + 14.758 \times 50}{110} = 7.25 \text{ kV}$$

$$\delta U_2 = \frac{P_2' X_L - Q_2' R_L}{U_2} = \frac{20 \times 50 - 14.758 \times 3}{110} = 8.69 \text{ kV}$$

$$\dot{U}_1 = \dot{U}_2 + (\Delta U_2 + j\delta U_2) = 117.57 \angle 4.23^\circ \text{ kV}$$

$$\begin{aligned}\Delta \tilde{S}_{Y1} &= -j\frac{1}{2}BU_1^2 = -j2 \times 10^{-5} \times 117.57^2 \\ &= -j0.276 \text{ MVar}\end{aligned}$$

$$\tilde{S}_1 = \tilde{S}'_1 + \Delta \tilde{S}_{Y1} = 20.153 + j17.035 \text{ MVA}$$

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